

A Survey of Large Language Models

Literature Review & Technical Breakdown of Zhao et al. (2023)

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Introduction & Background

Overview & Key Dimensions

- **Reference:** *A Survey of Large Language Models* (Wayne Xin Zhao et al., 2023, arXiv:2303.18223).
- **Scope:** Systematic review covering 140+ pages on LLM lifecycle.
- **Four Core Pillars:**
 1. From Language Modeling to Large Language Models
 2. Pre-training (Architecture, Data, Scaling Laws)
 3. Adaptation (Instruction Tuning, Alignment, PEFT)
 4. Utilization (In-Context Learning, CoT, RAG)
 5. Capability Evaluation (Reasoning, Hallucinations, Benchmarks)

From Language Modeling to Large Language Models

From Language Modeling (LM) to Large Language Models

“The limits of my language mean the limits of my world.” — Ludwig Wittgenstein

Language Modeling (LM) aims to model human language by predicting the next token in a sequence: $P(w_1, \dots, w_n) = \prod_{i=1}^n P(w_i \mid w_{<i})$.

The Four Historical Stages:

- **Statistical LM (before 1990s):** Focused on next-word prediction based on the Markov assumption and n -gram frequency statistics.
- **Neural LM (2012+):** Solving NLP tasks using distributed representations and deep learning architectures: Word2Vec, MLP, RNN, and LSTM.
- **Pre-Trained LM (PLM) (2017–2020):** Sparked by “Attention Is All You Need” (Vaswani et al., 2017). Introduction of the *pre-training* $\rightarrow$ *fine-tuning* paradigm (BERT, GPT-1, GPT-2) to address diverse NLP benchmarks.
- **Large Language Models (LLM) (2020+):** Paradigm shift initiated by GPT-3, followed by models like Claude and Gemini, scaling up parameters and compute to tackle general-purpose, real-world tasks via in-context learning.

Scaling Laws: Kaplan vs. Chinchilla

Cross-entropy loss L scales predictably as a power-law with parameters N , dataset size D , and compute budget C : $L(N, D) \approx \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D}$.

Kaplan et al. (OpenAI, 2020)

- Suggested allocating more compute to parameters rather than tokens ($N \propto C^{0.73}$, $D \propto C^{0.27}$).
- Led to massive, undertrained models (e.g., GPT-3: 175B parameters on 300B tokens).
- Flaw: Fixed learning rate schedules and under-represented dataset scaling.

Chinchilla (Hoffmann et al., 2022)

- Compute-optimal training requires scaling N and D in equal proportions: $N \propto C^{0.5}$, $D \propto C^{0.5}$.
- Optimal ratio:
 ≈ 20 tokens per parameter.
- Result: A smaller model trained on more data (Chinchilla: 70B on 1.4T tokens) outperforms GPT-3 and reduces inference costs.

Core Emergent Abilities

Abilities not present in smaller models that sharply emerge once scale crosses a compute/parameter threshold (Wei et al., 2022).

- **In-Context Learning (ICL):**

- Performing downstream tasks given only natural language instructions and a few demonstration examples $(x_1, y_1), \dots, (x_k, y_k)$ at test time.
- No gradient updates or parameter modifications; relies on internal attention dynamics (e.g., induction heads).

- **Instruction Following:**

- Generalizing to unseen task formats expressed in explicit human instructions.
- Unlocked via Supervised Fine-Tuning (SFT) over diverse task mixtures (e.g., FLAN, InstructGPT).

- **Step-by-Step Reasoning:**

- Standard prompting fails on complex multi-step problems (math, symbolic logic).
- Unlocked by Chain-of-Thought (CoT) prompting: generating intermediate reasoning paths before the final answer ($Q \rightarrow \text{Rationales} \rightarrow A$).

Public Datasets and Access Paradigms

Pre-training Corpora

- **Scale:** Trillions of tokens.
- **Web data:** Common Crawl, RefinedWeb, FineWeb.
- **Curated domains:** The Pile (PubMed, arXiv, StackExchange), Wikipedia, GitHub.
- **Focus:** Aggressive deduplication, heuristic filtering, perplexity scoring.

Fine-tuning & Alignment Data

- **Scale:** Thousands to low millions of samples.
- **Instruction datasets:** FLAN, Dolly, OpenAssistant, ShareGPT, UltraFeedback.
- **Focus:** Diversity, response quality, and multi-turn conversational flow.

Deployment & Consumption Paradigms

- **Open-weights:** Downloadable checkpoints for local hosting and full parameter tuning (LLaMA, Mistral, Qwen).
- **Commercial APIs:** Black-box inference access via token streaming endpoints; fine-tuning limited to adapter endpoints or LoRA upload (OpenAI API, Anthropic, Google Vertex AI).

Pre-training & Scaling Laws

Architectural Choices:

- Dominance of Decoder-only backbones (GPT, LLaMA).
- Positional Embeddings: RoPE vs ALiBi.
- Normalization: Pre-RMSNorm over Post-LayerNorm.
- Activation: SwiGLU.

Takeaways & Personal Critique

- **Trade-off:** Pre-training costs mandate parameter-efficient methods (PEFT) and retrieval augmentation (RAG) for applied engineering.
- **Key bottleneck:** Alignment and reasoning fidelity remain open empirical challenges rather than solved theoretical problems.